Vision-Based Control of Aerial Robotic System

A Project Report Submitted by

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Declaration

I hereby declare that the work presented in this Project Report titled Vision-Based Control of Aerial Robotic System submitted to the Indian Institute of Technology Jodhpur in partial fulfilment of the requirements for the award of the degree of Master of Technology, is a bonafide record of the research work carried out under the supervision of Dr. Anoop Jain. The contents of this Project Report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Signature

Kothawale Chetan Dattatraya

M24IRM003

Certificate

This is to certify that the Project Report titled Title of the Project Report, submitted by Kothawale Chetan Dattatraya(M24IRM003) to the Indian Institute of Technology Jodhpur for the award of the degree of Master of Technology, is a bonafide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature

Dr. Anoop Jain

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Abstract

Autonomous precision landing is very important for UAVs, especially on small, mobile, or dynamic surfaces. This work tries to develop and evaluate vision-based control strategies for a quadrotor to land autonomously on a moving platform. It uses a downward-facing camera to detect the platform without markers, relying on colour and shape features for real-time relative positioning without external sensors. A dual-loop control system is implemented here, the inner loop stabilises the UAV with PID control, while the outer loop manages vision-guided tracking and landing.

During the mid-term phase of the thesis, two controllers were designed and implemented in a Matlab + Python simulation environment using the Parrot Minidrone model (1) vision-based PID controller and (2) vision-based Lyapunov (Control Lyapunov Function) controller. Both controllers were tested for autonomous landing on a platform moving along a circular trajectory. Simulation results show that while the PID controller can achieve a successful landing under nominal conditions, it suffers from oscillations and higher control effort, and the Lyapunov controller demonstrates smoother convergence, improved robustness and higher landing precision due to its stability guarantees.

This work establishes the foundational components—vision perception, control architecture, and simulation framework—required for advanced learning-based approaches. The future work includes integrating a learning based controller for the outer loop to handle complex dynamics and uncertainties with real-world implementation on a physical drone platform.

Contents

Abstract **vi**

1 Introduction **2**

2 Literature survey **3**

3 Problem statement and Objective **5**

4 Methodology **6**

4.1 Mathematical Modeling of Quadcopter Dynamics 6

4.1.1 Translational Dynamics 6

4.1.2 Rotational Dynamics 6

4.2 Control Strategies 7

4.2.1 Vision-Based PID Controller 7

4.2.2 Control Lyapunov Function (CLF) 8

5 Simulation & Results **10**

5.1 Simulation Setup 10

5.2 Result 10

5.3 Performance Discussion 12

6 Conclusion **13**

7 Future plan of work **14**

References **15**

List of Figures

1.1 Quadrotor autonomous landing 2
4.1 Dynamics of the quadrotor 7
4.2 Block diagram of the quadcopter control 8
5.1 Platform detection using camera 10
5.2 PID controller performance analysis 11
5.3 Lyapunov-based controller performance analysis 11

List of Tables

5.1 Performance comparison of the two controllers in simulation. 12

Vision-Based Control of Aerial Robotic System

1 Introduction

Unmanned Aerial Vehicles (UAVs) or drones, are increasingly used for tasks like delivering packages, inspecting buildings, helping with transportation, and conducting search and rescue missions. One of the important features that makes these tasks possible is the ability to land automatically, especially in small, moving, or unpredictable areas. Landing on a moving platform can be difficult due to the complex nature of drones, environmental factors, and delays in how they see and interact with the surroundings.

This thesis develops a system that allows a quadrotor drone to land autonomously on a moving ground platform. The drone uses a downward facing camera to find the landing spot by recognising its colour and shape. Once it identifies the landing area, the drone starts to track it and decreases its altitude in a controlled manner to land accurately. A hierarchical control architecture is:

- Outer loop: Generates position tracking and landing commands based on vision data.
- Inner loop: Stabilises attitude and thrust using a classical PID controller.

During the mid-term evaluation, my focus was on understanding the mathematical modelling of the quadrotor, studying the interaction between inner and outer loops and implementing two fundamental control strategies:

- Vision-based PID Controller
- Vision-based Lyapunov Controller (CLF-based)

I was using MATLAB and Python simulation tools to implement and test both algorithms on a Parrot Minidrone model. In the simulation, the drone finds and follows the platform moving in a circular path before landing on it autonomously.

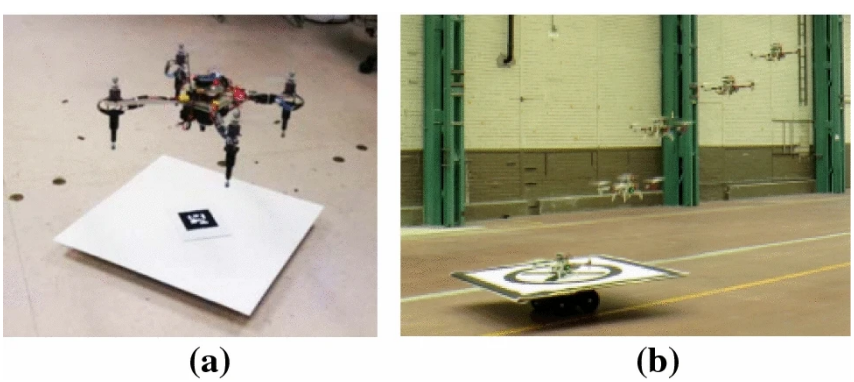

Figure 1.1: Quadrotor autonomous landing
[1]

2 Literature survey

Autonomous precision landing is important for quadrotors used in delivery, search and rescue, and returning safely after flights, especially in areas without GPS or in changing environments. As technology for aerial robots grows in cities and remote places, precise landing makes missions safer, saves energy, and improves reliability. This is especially important when quadrotors need to return to charging stations or work with ground vehicles. The ability to find and assess safe landing spots in complex or unexpected situations is key to the system's strength. This feature allows quadrotors to deal with risks from low batteries, system failures, or completing their tasks independently.[2]

- **Vision-Based Navigation for Landing:**

  Vision-based navigation enables drones to function effectively without relying solely on GPS or other external sensors. This is particularly beneficial in busy or cluttered environments where GPS signals may be unreliable. Quadrotors utilize either monocular or stereo cameras to perceive their surroundings. They gather visual data to determine their location, avoid obstacles, and identify safe landing spots. Modern techniques employ methods such as semantic segmentation, object detection, and mapping the environment in 2D or 3D. These systems integrate visual information with onboard planning, allowing them to adjust flight paths as needed. They can find suitable landing areas, anticipate moving obstacles, and guide the drone to land with precision. [3, 2, **?**]

- **From Marker-Based to Marker-Free Vision Control:**

  Traditional landing systems use visual markers like AprilTags or ArUco markers on the landing pad. These markers help the drone to detect and find the landing spots accurately at different distances and angles. However, these systems struggle in unpredictable environments and need markers to be placed in advance, which limits their use. [4]

  Recent developments aim to find landing sites without using artificial markers. Instead of that, they are using features in the environment. Key techniques include:

  1. Feature Extraction: New methods find important landmarks and patterns in images to help with navigation. These systems can track your position and speed using features from images or movement, enabling guidance even on uneven or moving surfaces. [5, 3]

  2. Contour Detection: Algorithms like neural networks and traditional computer vision methods can find shapes in the environment, including edges of rooftops or flat areas for landing. These techniques can quickly evaluate potential landing zones based on their shape and appearance.[2]

  Using real-world features allows these systems to work better in unknown or changing environments, making them more practical and reliable.

- **Recent Progress and Challenges:**

- We use both 2D and 3D information to find and evaluate safe landing sites, especially in messy or complicated areas.[2]
- Learning-based controllers can estimate where targets are located and adjust for landing pads that move or are uncertain.
- Planning movement and avoiding obstacles using vision to predict moving objects is important for navigating urban areas and complex environments. [6]
- Use model-free control and dynamic optimisation to create smooth landing paths. This approach helps improve stability when facing disturbances.[5]

A major challenge is understanding complex and unclear environments without using technology, especially when lighting, size, and shape change. Currently, researchers are working on using reinforcement learning, probabilistic estimation, and real-time planning to make landing controllers that don't rely on markers more adaptable and effective.

3 Problem statement and Objective

To design and implement a vision-based autonomous landing system for a quadrotor to land on a mobile platform moving in a circular trajectory.

- **Objectives:**

  1. To model the dynamic equations of a quadrotor and verify their accuracy through simulation.
  2. To design and implement a PID controller for stable attitude and position control.
  3. To detect and track the platform using a downward-facing camera and generate position commands based on image-plane errors.
  4. To Stabilise and guide the drone to perform a smooth and precise landing.
  5. To conduct a comparative study of the performance of PID and Lyapunov controllers.

4 Methodology

4.1 Mathematical Modeling of Quadcopter Dynamics

[7]

The quadrotor is modeled as a rigid body with six degrees of freedom (DOF), defined by three translational and three rotational motions. The nonlinear dynamics are derived from the Newton–Euler formulation. The model can be separated into translational and rotational components as follows:

4.1.1 Translational Dynamics

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \frac{1}{m} R(\phi, \theta, \psi) \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$$

where:

- x, y, z : position of the quadrotor in the inertial (world) frame,
- $\ddot{x}, \ddot{y}, \ddot{z}$: linear accelerations,
- m : total mass of the quadrotor (kg),
- g : acceleration due to gravity (9.81 m/s^2),
- T : total upward thrust generated by all four rotors (N),
- $R(\phi, \theta, \psi)$: rotation matrix transforming forces from the body frame to the inertial frame.

The rotation matrix is defined as:

$$R(\phi, \theta, \psi) = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \sin\phi\cos\theta \\ \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi & \cos\phi\cos\theta \end{bmatrix}$$

4.1.2 Rotational Dynamics

The rotational motion is governed by:

$$I\dot{\omega} = \tau - \omega \times (I\omega) \tag{1}$$

where:

- $I = diag(I_x, I_y, I_z)$: moment of inertia matrix,
- $\omega = [p, q, r]^T$: angular velocity vector,
- $\tau = [\tau_\phi, \tau_\theta, \tau_\psi]^T$: torque vector generated by rotors.

The total thrust and control torques are derived from the individual motor thrusts as:

$$\begin{bmatrix} T \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ l & 0 & -l & 0 \\ k_m & -k_m & k_m & -k_m \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$$

where:

- $F_i = k_f \omega_i^2$: thrust generated by rotor i,
- l : distance between rotor and quadrotor center (m),
- k_f : thrust coefficient (N·s^2/rad^2),
- k_m : drag moment coefficient (N·m·s^2/rad^2).

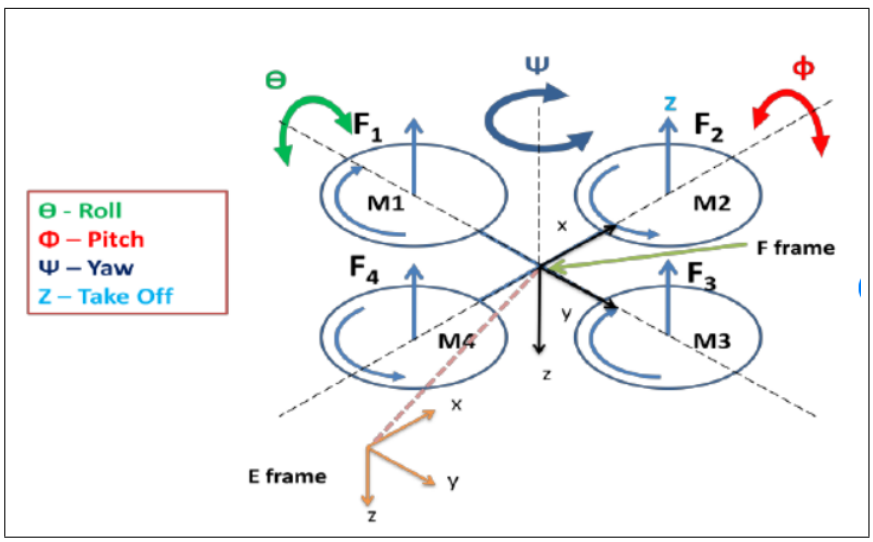

Figure 4.1: Dynamics of the quadrotor

4.2 Control Strategies

4.2.1 Vision-Based PID Controller

The Proportional–Integral–Derivative (PID) controller computes control inputs based on tracking error:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \tag{2}$$

where:

- $e(t) = r(t) - y(t)$: tracking error between desired (r) and measured (y) states,
- K_p : proportional gain, determining response speed,

- K_i : integral gain, reducing steady-state error,
- K_d : derivative gain, reducing overshoot and oscillations.

A nested control loop structure was implemented:

1. **Rate Control (Inner loop):** Controls angular rates (p, q, r),
2. **Attitude Control (Middle loop):** Controls angles (ϕ, θ, ψ),
3. **Position Control (Outer loop):** Controls translational motion along (x, y, z) using desired attitude angles.

Each loop operates at different frequencies the inner loop running faster for stabilization and the outer loop running slower for position tracking. [3]

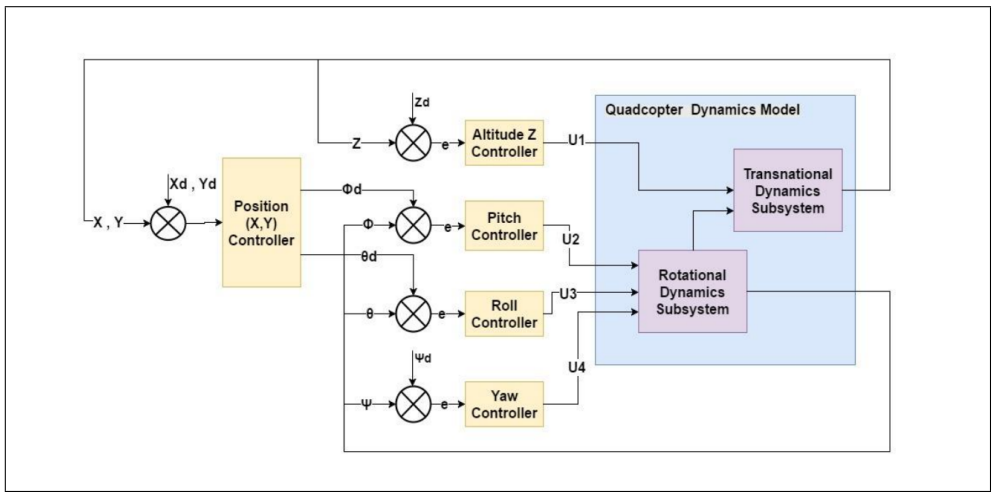

Figure 4.2: Block diagram of the quadcopter control
[8]

4.2.2 Control Lyapunov Function (CLF)

Lyapunov function: [6]

$$V = \frac{1}{2}(e_x^2 + e_y^2)$$

To ensure stability:

$$\dot{V} = e_x \dot{e}_x + e_y \dot{e}_y < 0$$

Control law:

$$u_x = -ke_x, \qquad u_y = -ke_y$$

where,

- V : Lyapunov function measuring system energy/error
- $\dot{V}$: Time derivative of Lyapunov function
- k : Positive control gain

5 Simulation & Results

5.1 Simulation Setup

- Simulation Software: MATLAB/Simulink + Pybullet(Python)
- Drone: Parrot Minidrone model
- Camera: Downward-facing RGB camera
- Platform motion: Circular trajectory on ground
- Task sequence:
    1. Takeoff to fixed altitude
    2. Detect platform
    3. Track platform motion
    4. Controlled descent
    5. Landing on the moving platform

5.2 Result

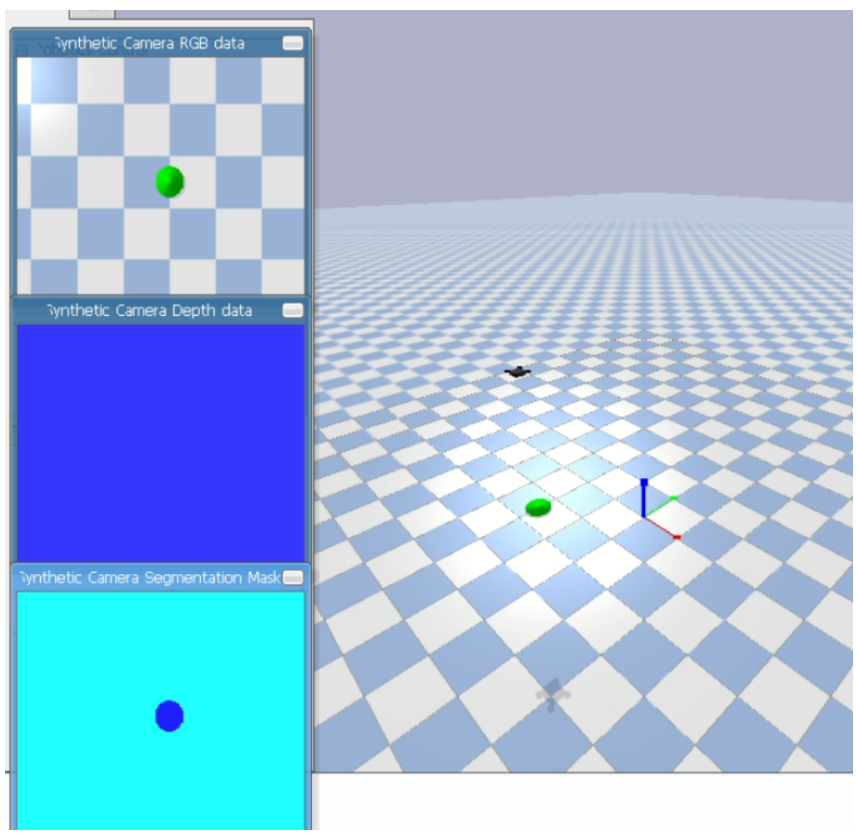

Figure 5.1: Platform detection using camera

The simulation results show that we can use vision-based control for autonomous landing. Both controllers are able to reduce the position error to zero, allowing drone to land on the circular mobile platform. This result confirmed that we can navigate using position errors comming directly from a camera.

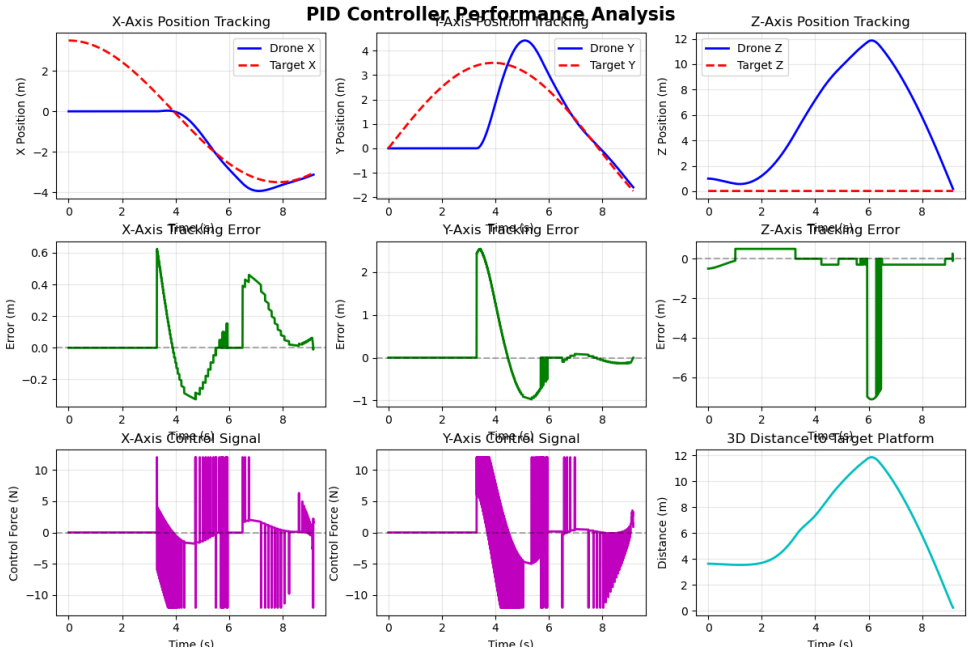

Figure 5.2: PID controller performance analysis

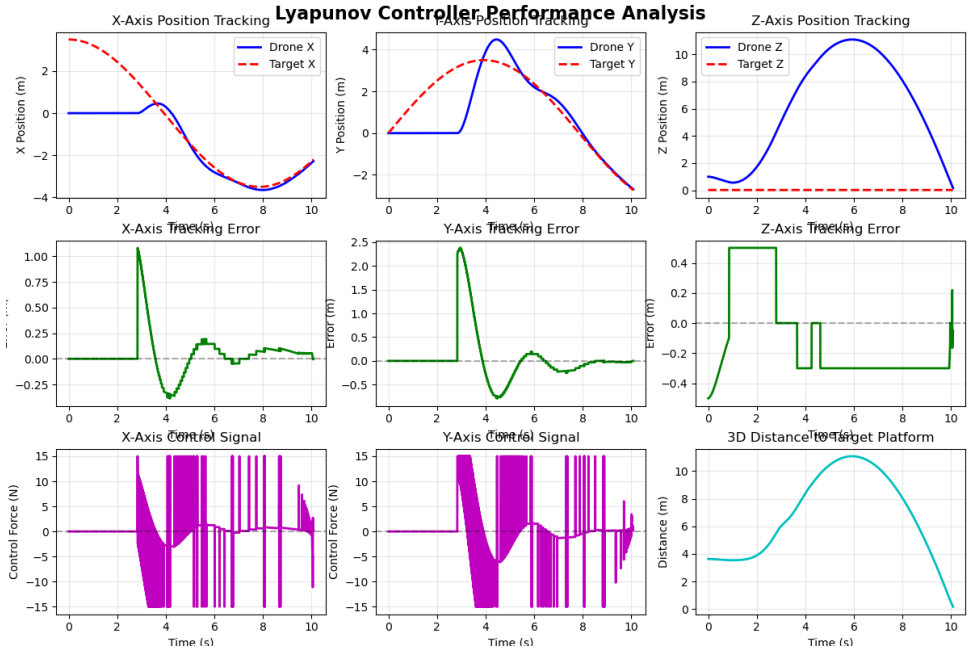

Figure 5.3: Lyapunov-based controller performance analysis

5.3 Performance Discussion

Table 5.1 summary of performance matrices of PID and CLF controllers.

Metric	PID Controller	Lyapunov Controller
Tracking accuracy	Moderate	High
Control smoothness	Low	High
Landing precision	Acceptable	Very high
Settling time	Longer	Shorter
Robustness	Low	High
Control effort	High oscillation	Smooth and minimal
Overall performance	Medium	Superior

Table 5.1: Performance comparison of the two controllers in simulation.

- **Tracking Accuracy:** Lyapunov-based control helps the drone to reduce tracking errors to settle quickly. In comparison, PID control shows overshoot and oscillations.
- **Stability:** Due to its Lyapunov formulation, the CLF controller give guarantees of stabilisation, and smooth, reliable performance.
- **Control Smoothness:** The PID controller reacts more aggressively to the error, resulting in higher control energy and potential instability under fast target motion.
- **Landing Precision:** The Lyapunov controller shows consistently landing closer to the platform centre.
- **Settling Time:** The Lyapunov controller reduces overall landing time by achieving error convergence faster.

6 Conclusion

During the mid-term phase, we successfully developed and tested two autonomous landing algorithms: one using a vision-based PID controller and another using a vision-based Lyapunov controller. We used MATLAB/Simulink with a Parrot Minidrone model. This work involved creating a model of the quadrotor, using a downward-facing camera for perception, designing outer-loop controllers, and combining them with an inner-loop PID stabilizer.

The simulation results showed that although the PID controller is simple, the Lyapunov-based controller is much better in terms of stability, robustness, and landing accuracy. This work lays a strong foundation for the next phase of the thesis.

7 Future plan of work

The final phase will focus on integrating learned intelligence and transferring the system to hardware:

- Development of a Vision + Reinforcement Learning based controller for outer-loop landing decisions.
- Hybrid architecture combining RL with PID/CLF inner-loop stabilisation.
- Improved vision module using advanced detection and prediction techniques.
- Transition to ROS2/Gazebo or MATLAB hardware-in-the-loop (HIL) simulation.
- Real hardware implementation and testing with onboard camera.
- Detailed comparison of simulation vs. hardware results.

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